Experimental Methodology

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Agenda

1. Affect Elicitation
2. Experimental Methodology
    1. Institutional Review Board (IRB)
3. Research and Development Tools
    1. Demo: PsychoPy

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Affect Elicitation

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Dataset of emotional expressions

1. Acted or posed Expressions obtained by asking individuals, often actors, to portray emotions
    i. Easy to collect
    ii. Ecological validity of data is a concern
2. Naturalistic display of emotions
    i. Ecologically valid
    ii. Notoriously difficult to collect
3. Induced Expressions, in which emotional responses are elicited via some stimulus;
    i. Middle road

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Emotion Elicitation

1. Passive or perception based
   - Individuals observe stimuli, such as film clips, images or music that are designed to evoke particular feelings or methods

2. Active or Expression based
   - Individuals are instructed to perform particular behaviours that might naturally evoke different emotions, such as posing facial muscles, adopting body postures or interacting with other people.

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Passive Methods: Images

1. The presentation method is standardized such that all the individuals have the same viewing experience;
   - E.g. the images could be presented for 10 seconds on a computer screen that is at a fixed distance; with a constant screen resolution, screen brightness, and image size (Monkaresi et. al, 2012).

2. The images can be selected on the basis of which emotion should be elicited from a database of standard images.
   - International Affective Picture System (IAPS)
   - Genova Affective Picture Database (GAPED)

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Image Databases

Source - Emotional Design: International affective picture system (IAPS)

Passive Methods: Images

a. Advantages:
   i. Noninvasive
   ii. Easily accessible
   iii. Relatively simple setup
b. Disadvantages:
   i. Strength of emotion is lower
   ii. Emotional reactions are short and transient
   iii. Lack of Personalization:
      1. Individuals with cocaine addiction react differently to pleasant, unpleasant and neutral IAPS images (Asensio et. al, 2010)
   iv. Etc.
c. Utilizing images may be useful for reactive modalities such as facial expressions, physiological signals but not for productive modalities (text or gestures etc.)

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Passive Methods: Film Clips

1. For comparison purposes, a neutral baseline film is generally shown prior to the presentation of each emotional clip.
   - neutral clip [12 seconds] -> emotional clip [1-2 minute] -> self-assessment phase
2. Short clips or long clips?
3. Physical situation should be standardized (Rottenberg et. al, 2007):
   - E.g. 20 inch monitor about 5 feet from the participant
4. For each target emotion, 1 to 2 short clips as homogenous as possible
5. The flims can be selected online from various sources, e.g:
   - https://sites.uclouvain.be/ipsp/FilmStim/

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Passive Methods: Sample

Source - https://sites.uclouvain.be/ipsp/FilmStim/En/2.mp4

Passive Methods: Film Clips

1. Advantages:
   a. They capture the attention well
   b. Can elicit higher intensity emotions and more complex emotional states (Rottenberg et. al, 2007)
   c. Also useful in studying the emotion latency, rise time , duration and offset
2. Disadvantages:
   a. Ecological validity is in question
   b. Individuals may have seen the films from which the clips are taken
   c. Not ideal for obtaining data from productive modalities.
      - Gestures or verbal behaviour

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Active Methods:Behavioral Manipulation

1. Individuals are instructed to adopt particular muscular configurations or behavioral patterns that have been associated with emotional expressions, such as contracting or relaxing facial muscles or exaggerating natural emotional expressions (Ekman et. al, 1993, 2007).

2. Directed Facial Action Task (Ekman et. al, 2007)
   a. Method for eliciting emotion through the manipulation of facial expressions

3. Researchers often ask participants to think or write about an instance(s) from their past when they experienced an emotion such as anger (e.g., Siedlecka et al., 2015).

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Active Methods: Facial Action Task

Source: Dhruv Verma, Sejal Bhalla, Dhruv Sahnan, Jainendra Shukla, and Aman Parnami. 2021. *ExpressEar: Sensing Fine-Grained Facial Expressions with Earables*. Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 5, 3, Article 129 (Sept 2021), 28 pages.

Active Methods:Behavioral Manipulation

1. Disadvantages:
   a. Ecological validity: Can one really say that the emotions elicited through behavioural manipulation are more “pure”?
   b. One needs to know the physical behaviour associated with target emotions.
      i. Ekman's work does not include more complex emotions such as frustration, confusion, engagement etc.
      ii. Limits the range of target emotions
   c. Poses or actions required can be difficult or complicated.
2. Advantages:
   a. Useful for collecting reactive expressions
   b. The effects are strong if physical behaviour associated with target emotions are known precisely.

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Active Methods: Social Interaction

- Advantages:
  a. Elicited emotions are more realistic and natural.
  b. Easy to elicit emotions that are otherwise hard to elicit, such as anger, guilt.

Source: https://brewminate.com/wp-content/uploads/2018/02/022818-122-Sociology-Social-Interaction.jpg

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Experimental Methodology

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IRB for Human Research

- Research Involving Human Subjects: Institutional Review Board (IRB)
- IRB is a review committee established to help protect the rights and welfare of human research subjects.
- Documents for IRB:
  - A draft or an abstract of the proposal
    - A clear description of the research methodology, or the experimental design.
  - A copy of the "informed" consent form(s).
  - A description of how confidentiality/anonymity will be protected.
  - A description of the risks and benefits to the subjects.
  - A copy of any recruitment document (advertisement flyers/invitation letters/invitation emails).
  - Data collection instruments the survey, or sensors, description of the experimental research method, or the list of questions or interview protocols.

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Criteria for IRB Approval

1. Risks to subjects are minimized or reasonable.
2. Selection of subjects is equitable.
3. Informed consent is sought.
4. Plans for monitoring the data collected.
5. Adequate provisions to protect the privacy of subjects and to maintain the confidentiality of data.
6. Additional safeguards to protect vulnerable population.

Source: https://i.imgflip.com/36usb7.jpg

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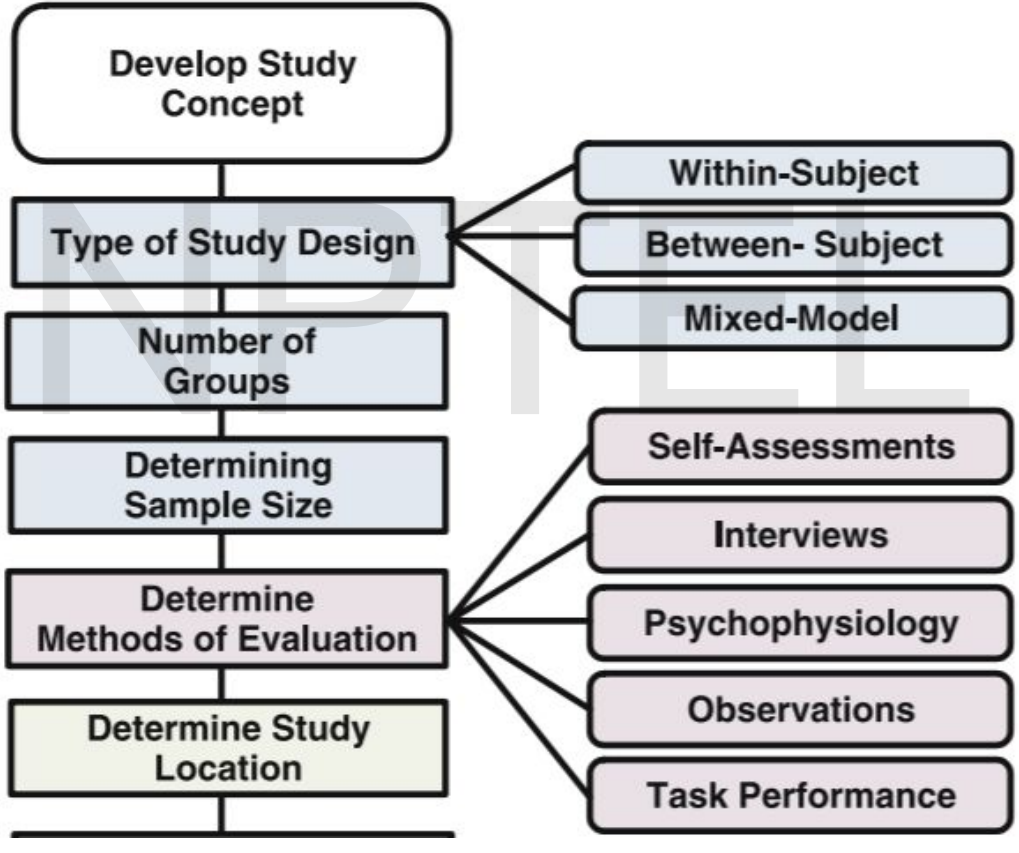
Experimental Design - I



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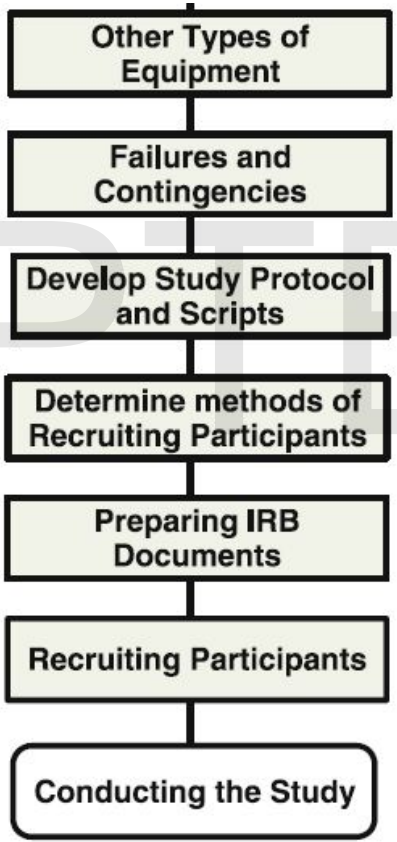
Experimental Design - II



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Research and Development Tools

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Category of Tools

- Data Collection
- Data Annotation
- Signal Analysis
- Affect Classification
- Affect Expression

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Data Collection

- PsychoPy
- OpenSesame
- Experiment Wizard
- SuperLab
- E-Prime
- Presentation
- DMDX
- Paradigm
- In-house

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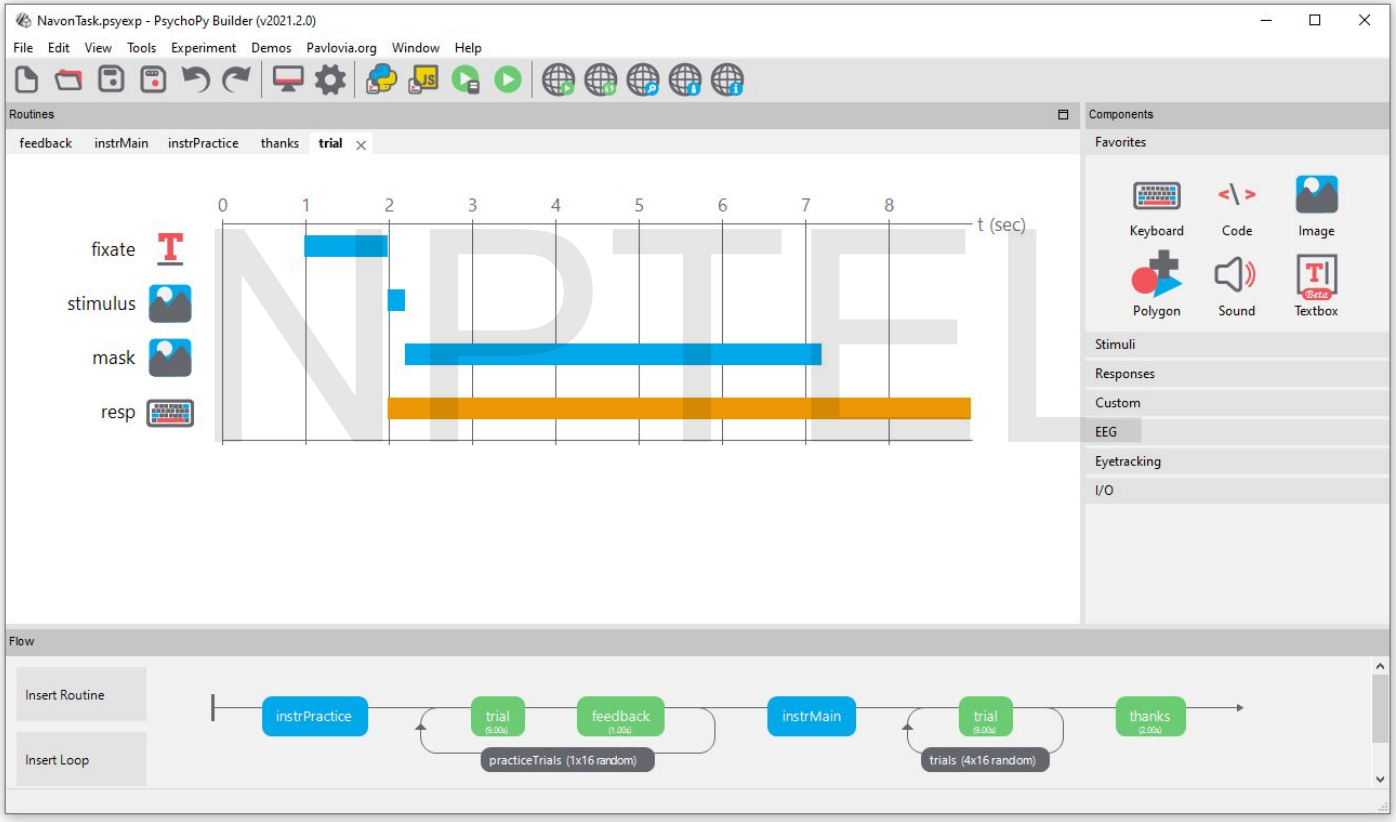
Psychopy



Source: https://www.psychopy.org/

Labelling and Data Annotation

- Genova Emo Wheel
- Self-Assessment Mankins (SAM)
- Feeltrace
- Tracetools
- Audio
  - ELAN
  - Wavesurfer
  - Praat
  - Speechalyzer
- Video:
  - ELAN
  - Cowlog
  - ANVIL
  - Gtrace
  - ChronoViz
- Text
  - Whissell Dictionary of Affect in Language
- In-house

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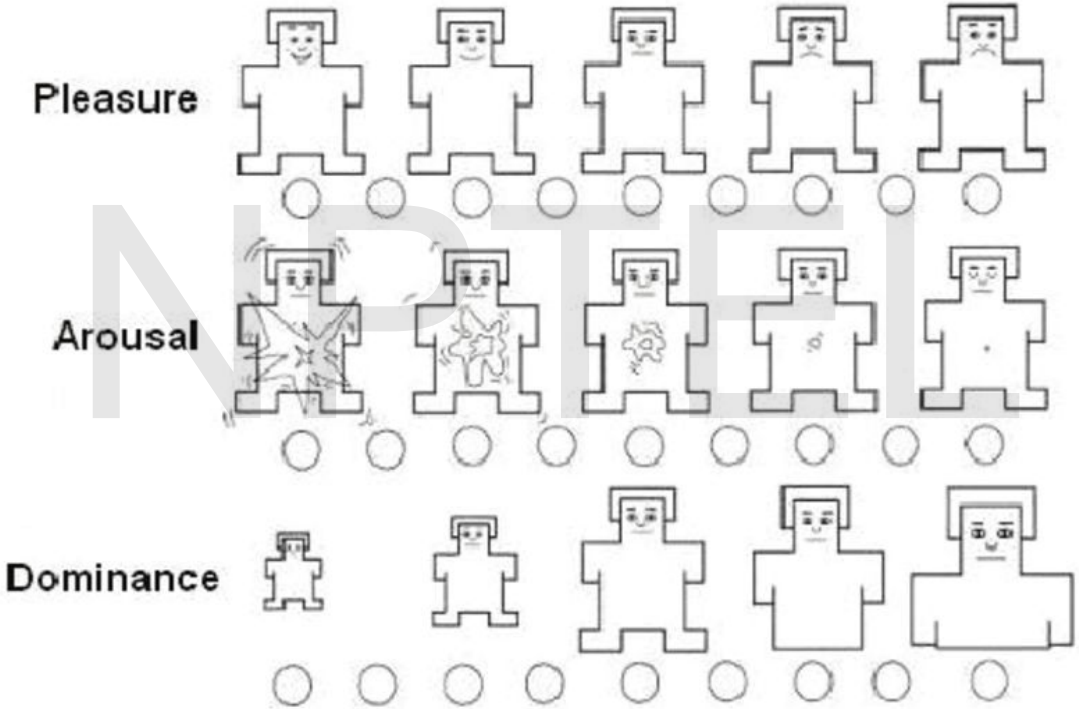
SAM Mankins



Source: Morris, J.D. (1995). Observations: SAM: The Self-Assessment Manikin An Efficient Cross-Cultural Measurement Of Emotional Response 1. Journal of Advertising Research.

Signal Processing and Analysis

- Audio:
  - Praat
  - OpenEAR
  - OpenSmile
  - Wavesurfer
- Physiological Signals
  - EEGLab, PyEEG
  - ELAN
  - AuBT
- Image/Video
  - OpenCV, OpenPose
  - Kinect SDK
  - MediaPipe
- In-house

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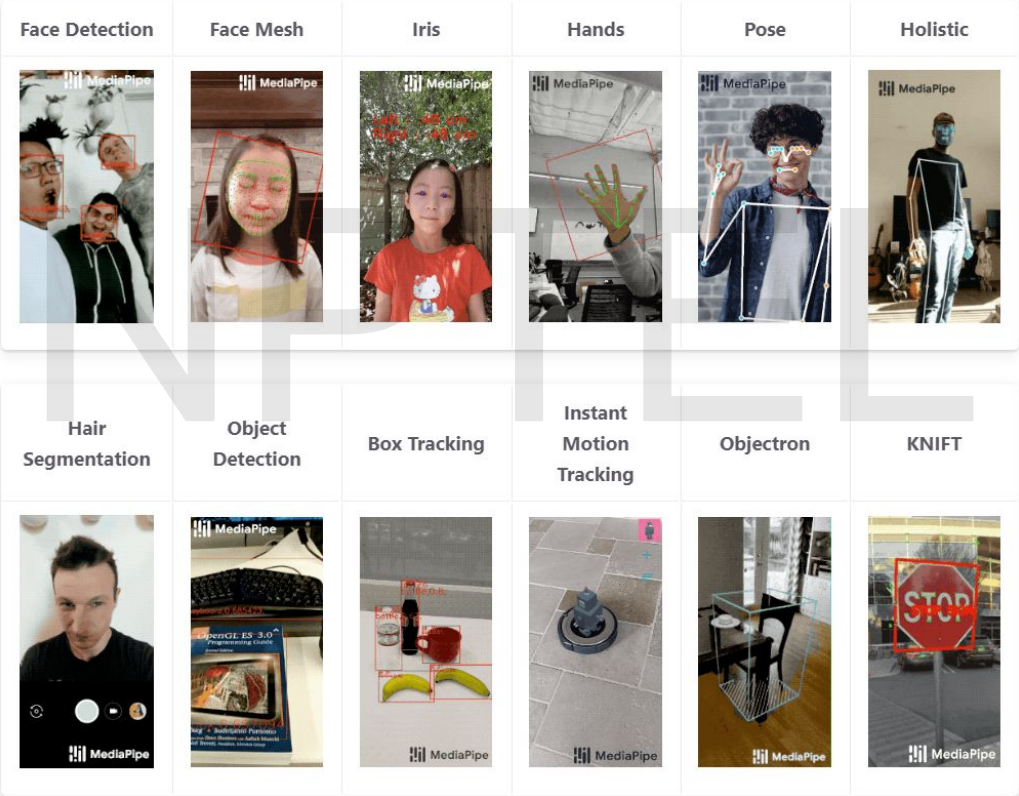
MediaPipe



Source: https://google.github.io/mediapipe/

Data Mining

- WEKA
- AutoML
- SPSS
- PRTools
- MATLAB Arsenal
- R
- LibSVM, SVMlight
- HTK
- SAS
- RapidMiner
- SciPy

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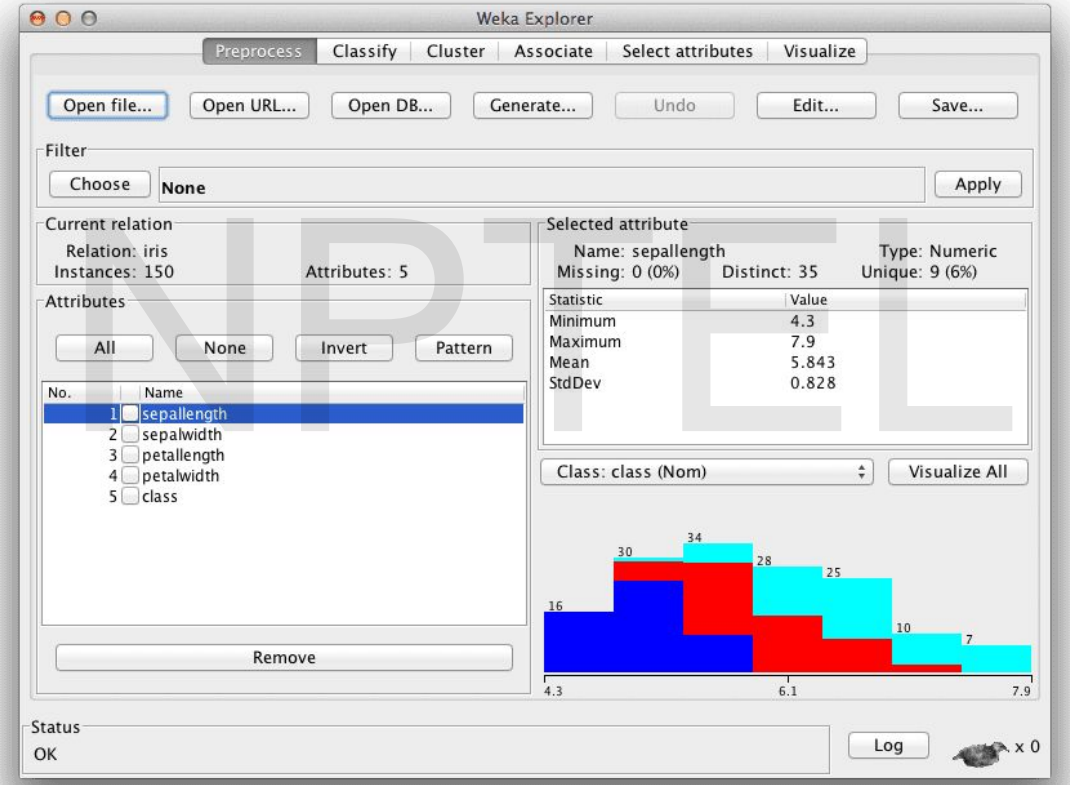
WEKA



Source: https://machinelearningmastery.com/wp-content/uploads/2014/02/weka-explorer.png

Affect Expression

- MARY
- GRETA
- Festival
- VHML
- SmartBody
- FaceFX
- Xface
- Horde3D
- Haptek
- ROS
- ICT Virtual Human Toolkit
- SOAR
- ACT-R

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ICT Virtual Human Toolkit

Source: https://vhtoolkit.ict.usc.edu/

References

- http://people.ict.usc.edu/~gratch/CSCI534/Lecture2020-04-methods%20review.pdf
- Research Methods in Psychology
- Review of Human Studies Methods in HRI and Recommendations
- Chapter 27, OHAC
- Development and validation of Image Stimuli for Emotion Elicitation (ISEE): A novel affective pictorial system with test-retest repeatability
- It's the Effect Size, Stupid: What effect size is and why it is important

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